1.3 · Agent Use Cases and Workflow Mapping

Week 2 · Monday Jul 20 — LLMs and Agentic AI Workflows

Generative AI & Sustainability · Munich Summer School 2026

Outline

- Compare good and bad examples of agentic workflows
- Discuss the limits of LLM outputs and the need for human review
- Review possible agents for the learning space
- Compare learning concierge, group formation, room adaptation, sustainability feedback, project coach, accessibility, and facility agents
- Introduce the agent-map structure: user input, decision, tools/data, output, human approval, and failure mode

Where we are

- 1.2 gave you the parts: **loop + tools + memory + planning + routing + human gate**
- This session: how to **assemble** them well — and how assemblies go wrong
- This afternoon your team commits to **three agents** and drafts the **workflow diagram** your final pitch depends on

Good vs. bad agentic workflows

The same building blocks can produce a helpful assistant or an expensive surveillance machine.

What a *good* agentic workflow looks like

- **Bounded scope** — does one job; you can say what it will *never* do
- **Verifiable output** — a person can check the suggestion before it matters
- **Right-sized autonomy** — the model decides only where flexibility helps
- **Right-sized model** — small/local where possible, frontier only where needed
- **Honest failure** — when unsure, it says so and hands off to a human
- **Minimal data** — touches only the data its job requires

Good example: sustainability feedback agent

- **Observes:** room energy meter + AI-usage log (counts only, no content)
- **Reasons:** usage this week vs. last; which activities drove it
- **Acts:** posts a weekly digest — "chat queries doubled; try the cached course FAQ" — to the room dashboard
- Why it's good:
  - read-only tools, **can't change anything**
  - aggregated data, no individual tracking
  - wrong output = a mildly wrong message, **not a harmed person**

Bad example: same goal, poor design

"EcoEnforcer" — an agent that monitors each student's AI usage, ranks students by carbon footprint, and blocks the "worst offenders" from the room's AI tools

- **Individual productivity scoring** — high-risk data, corrosive to trust
- **Automated punishment** — irreversible action, no human gate
- **Unbounded inference** — burns energy monitoring everyone to save energy
- Same mission as the previous slide — **design choices** made it harmful

How agent workflows fail

- **Hallucinated actions** — cites a room that doesn't exist, invents a policy
- **Runaway loops** — retries forever; cost and energy climb silently
- **Tool misuse** — right tool, wrong arguments: books the wrong room for the wrong week
- **Prompt injection** — text the agent *reads* (an email, a web page) hijacks what it *does*
- **Scope creep** — helpful assistant slowly becomes attendance tracker
- **Silent failure** — worst of all: wrong, confident, and nobody's checking

Why human review is non-negotiable

- An LLM's output is a plausible guess — there is no built-in guarantee of truth, and no reliable "I'm sure" signal
- So place review where the stakes are:
  - low stakes, reversible → let it run (draft text, suggestions)
  - visible to others → spot-check (dashboards, digests)
  - about people or irreversible → explicit approval gate
- Q&A on Thursday will probe exactly this: "*what if your agent is wrong?*"

Agents for your learning space

Seven candidates from the challenge brief — your team picks at least three.

The candidate roster

Agent	One-line job
Learning concierge	finds materials, explains concepts, suggests next steps
Group formation	proposes project teams from skills and availability
Room adaptation	suggests room modes from occupancy, CO_2, noise, light
Sustainability feedback	shows energy/compute impact, suggests lighter options
Project coach	guides design steps, stakeholders, risks, pitch prep
Accessibility & inclusion	translation, simplified formats, speech-to-text
Facility & maintenance	detects issues, collects feedback, files tickets

Compare: what data does each need?

- **Lightest touch:** facility (room state), sustainability feedback (meters), room adaptation (environmental sensors — no identities)
- **Middle:** learning concierge, project coach — conversations; fine **if not stored** or mined
- **Heaviest:** group formation — skills, availability, *judgments about people*
- First filter for this afternoon: **can this agent do its job with data you'd happily see collected about yourself?**

Compare: where can each go wrong?

- Room adaptation — annoying if wrong (lights change mid-presentation); low harm *if suggestions, not automation*
- Learning concierge — hallucinated citations, outdated materials; needs retrieval + sources
- Group formation — discriminatory grouping, self-fulfilling labels; **needs a human gate + explainable criteria**
- Accessibility — mistranslation harms exactly the people it serves; keep a "flag this" loop
- Facility — false tickets waste staff time; cheap to verify

Choosing your three

- **Value** — which top-5 student need from Week 1 interviews does it serve?
- **Feasibility** — could you prototype one interaction of it this week?
- **Data risk** — lightest data that still delivers the value?
- **Coverage** — a portfolio: one *learning* agent, one *space* agent, one *sustainability or inclusion* agent tends to pitch well
- **Demo-ability** — Thursday you show **one concrete interaction**; pick at least one agent you can stage live

The agent map

One diagram per agent — six boxes. This is your workshop template.

Six boxes every agent map needs

Box	Question it answers
1 · User input	What arrives, from whom?
2 · Agent decision	What does the model decide or infer?
3 · Tools / data	What can it read? What can it call?
4 · Output	What comes out — suggestion, draft, action?
5 · Human approval	Who confirms, before what?
6 · Failure mode	Most likely wrong outcome — and who catches it

If you can't fill a box, you don't have a design yet — you have a wish.

Worked example: room adaptation agent

Box	Entry
User input	team taps "workshop mode" on room panel
Agent decision	infers layout + light + air needs from mode, occupancy, CO_2
Tools / data	occupancy, noise sensors, light control API *(read + propose only)*
Output	"brighten front, TV notification, clear whiteboards"
Human approval	any person in the room confirms on the panel
Failure mode	wrong scene mid-use → one-tap undo; all changes reversible

Common mapping mistakes

- No failure box — "it just works" is not a design
- Human approval everywhere — approval fatigue; people click yes blindly; gate only what matters
- Tools listed as "the internet" — name the specific source or API
- Output box contains an essay — outputs should be small and checkable
- One agent doing three jobs — split it; three clear maps beat one blurry one

This afternoon: team agent-design workshop

- 1:15–2:45 · you will produce:
    - three selected agent use cases (with the choosing-criteria from today)
    - one agent map per agent — six boxes each
    - a first prototype idea — the one interaction you could demo
- Use today's building blocks: where does each agent use retrieval, tools, memory, routing?
- Instructors circulate — call us over early, not at 2:40

Wrap-up

- Good agents are **bounded, verifiable, honestly fallible** — bad agents are usually over-autonomous, over-collecting, or unsupervised
- Pick your three agents for **value, feasibility, data risk, and demo-ability**
- The six-box agent map is the contract between your idea and your pitch — Thursday's jury will look for boxes 5 and 6
- After lunch: **1.4 · Team Agent-Design Workshop** — bring the roster, leave with maps